

DATA 2010 Project Report: Trends in Olympic Athlete Performance and Representation (1896–2024)

Winter 2026

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1 Project Overview

Research question (what we are trying to find out). The Olympic Games provide a long-running record of athlete participation and medal outcomes, allowing us to study how performance and representation evolve over time. Using the *Olympic Summer and Winter Games (1896–2024)* dataset (athlete–event records with variables such as **year**, **season**, **city**, **sports**, **event**, **team**, **sex**, **age**, and **medal**), this project investigates three connected questions. First, we examine how medal outcomes vary by **age group** and **sex**, and how **gender parity** and **typical medal-winning age** changed across decades. Second, we evaluate how Olympic medal tables change when we count medals *per athlete* versus *per event*, and which countries become more or less “medal-efficient” over time when we account for participation size. Third, we examine whether countries win medals more efficiently when they spread success across many sports or focus on a few sports, and how this relates to rankings over time.

Why it matters (who would care and why). The answers matter to stakeholders who make decisions about athlete development and sport investment. National sport organizations and coaches can use age- and sex-specific medal patterns to support talent pipeline planning (e.g., whether medal-winning ages trend younger or older, and how quickly parity improves). Analysts, journalists, and policymakers can benefit from fairer country comparisons by understanding how medal totals depend on the counting method (athlete-medal vs. event-medal), especially for team sports. Finally, understanding whether diversified medal port-

folios across sports are associated with stronger efficiency and more stable rankings can inform long-term national sport planning.

Main result . [Write after completing analysis: briefly summarize 2–4 key findings that directly answer the three sub-questions.]

2 Research Question (Make it Specific)

The Olympic Games are not only a global competition but also a long-running dataset that can reveal how representation, performance, and strategic investment in sport evolve over time. Using Olympic records spanning 1896–2024, this project investigates patterns with real-world relevance to athlete development and national sport planning. Specifically, we focus on three research questions:

Our 3 Sub-questions

- 1) How do medal outcomes vary by age group and sex, and how did gender parity and typical medal-winning ages change across decades?
- 2) How do Olympic medal tables change when we count medals per athlete versus per event, and which countries become more or less “medal-efficient” over time when we account for participation size?
- 3) Do countries win medals more efficiently when they spread success across many sports, or when they focus on a few sports—and how does this affect medal rankings over time?

3 Data Source and Description

3.1 Where the data came from

- **Dataset name:** Olympic Summer and Winter Games (1896–2024)
- **Link (Kaggle):** <https://www.kaggle.com/datasets/magicchris/olympic-summer-and-winter-games-1896-2024>
- **Dataset owner (as shown on Kaggle):** Cassini-Chris (Owner)
- **Authors:** Not specified on the Kaggle dataset page
- **Provenance:** Not specified on the Kaggle dataset page (no source/citation details provided).
- **Citations / DOI:** Not specified on the Kaggle dataset page.
- **What each row represents:** One row represents one athlete’s record in one Olympic event for a specific Olympic Games, including the athlete’s sex/age (if present), country/team, the sport and event, the Games year/season/city, and the medal outcome (if any).
- **Key columns used (2–8 columns):**
 - **year** (Olympic year)
 - **season** (Summer/Winter)
 - **team** (country/team represented)
 - **sports** (sport category)
 - **event** (specific event)
 - **sex** (athlete gender)
 - **age / age_imputed** (athlete age)
 - **medal** (Gold/Silver/Bronze/No medal)
- **License:** Apache License 2.0 (as listed on the Kaggle dataset page).
- **Expected update frequency:** Not specified on the Kaggle dataset page.

3.2 Basic dataset facts

Number of rows	290,697
Number of columns	8
Time range (Year)	1896–2024
Seasons included	Summer & Winter
Unique years	38
Editions (unique year+season+city)	3136
Unique teams (team)	1151
Unique sports (sports)	92
Unique events (event)	1534
Medalist rows (Gold/Silver/Bronze)	247,941
Age range (years)	10–72
Sex proportion (M/F)	69.38% / 30.62%
Missing values present?	No
Exact duplicate rows removed	0

4 Data Cleaning and Feature Engineering

This section documents how we prepared the Olympics dataset for analysis and created new variables to support the three research questions.

Minimum expectations (what applies here):

- Fixed data types (e.g., **year** and **age** converted to numeric).
- Handled missing values (imputed missing **age** using group medians).
- Removed duplicates / impossible values (removed exact duplicate rows; dropped rows with invalid **year**).
- Dealt with outliers *carefully* (flagged suspicious ages instead of deleting them).

What we actually did:

- Standardized column names (lowercase with underscores) and stripped whitespace from text fields (e.g., **team**, **sports**, **event**, **medal**).
- Converted **year** to numeric and removed rows with missing/invalid **year**, since time is required for all analyses.

- Converted `age` to numeric (invalid values coerced to missing) and examined the distribution for unusual values.
- Standardized `medal` values by filling missing medals as `No medal` and ensuring consistent labels for `Gold`, `Silver`, `Bronze`, and `No medal`.
- Removed exact duplicate rows to prevent double counting.
- Flagged suspicious ages using an indicator `age_flag_suspicious` (e.g., very young or unusually old ages) to avoid deleting outliers without justification.
- Imputed missing ages by filling `age` with the median age within each (`sex`, `sports`) group, with an overall median used as a fallback, and stored the result as `age_imputed`.
- Engineered new features:
 - `decade` = $\lfloor \text{year}/10 \rfloor \times 10$ to analyze long-term trends.
 - `age_group` (Olympic-style bins such as `<16`, `16--18`, `19--25`, ..., `51+`).
 - `won_medal` (binary indicator for Gold/Silver/Bronze).
 - `medal_points` (Gold=3, Silver=2, Bronze=1, No medal=0) for summary comparisons.
- Created an *event-medal* view for country-level analysis by deduplicating medals at (`year`, `event`, `team_clean`, `medal`). This reduces team-event inflation and makes medal-table comparisons more interpretable.

Show evidence (before vs. after):

Check	Before cleaning	After cleaning
<code>year</code> type	mixed / not guaranteed numeric	integer (<code>int</code>)
Missing <code>year</code> rows	present	removed
<code>age</code> type	mixed / non-numeric values	numeric (<code>float</code>)
Missing <code>age</code>	present	imputed (with median)
Exact duplicates	possible	removed
Medal labels	inconsistent / missing possible	standardized (<code>Gold</code> / <code>Silver</code> / <code>Bronze</code> / <code>No medal</code>)
Outliers	unclear	flagged (<code>age_flag_suspicious</code>)

5 Exploratory Data Analysis and Visualization

You should include **at least 4** well-labeled figures. Every figure must answer: *What should the reader learn from this? Tell a story about your data and the plot.*

Research Question 1: How do medal outcomes vary by age group and sex, and how did gender parity and typical medal-winning ages change across decades?

5.1 Plot 1: Medalist Share by Age Group (100% stacked by sex)

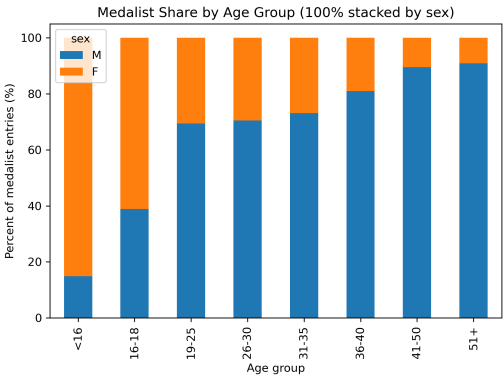


Figure 1: Each bar sums to 100%, showing the female vs. male share of medalist entries within each age group. This makes comparisons fair even when some age groups have far fewer medalist entries.

This figure indicates that sex balance among medalists varies by age group. Some age groups are more male-dominated, while others are closer to balanced. Differences are influenced

by sport mix (e.g., some sports peak younger or older) and historical differences in female participation.

Limitations: This (with `age_imputed`) does not account for age groups that can inflate counts because multiple athletes receive medals in a single event. Age imputation may introduce a smooth pattern in `age_flag_suspicious`.

5.2 Plot 2: Medal Type Share by Age Group (Gold/Silver/Bronze)

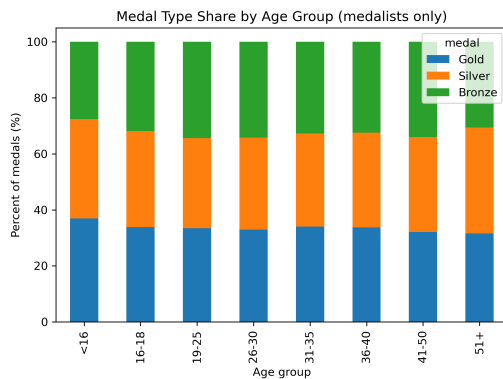


Figure 2: Each bar sums to 100%, showing how medals within each age group split into Gold, Silver, and Bronze. This focuses on the medal outcome mix rather than total volume.

This plot helps evaluate whether certain age groups have a relatively higher share of Gold compared to others. If the medal-type shares are similar across age groups, it suggests age affects the frequency of medals more than the distribution of medal types.

Limitations: Percent shares hide total medal volume; a small age group can still sum to 100%. Medal type is influenced by sport and competition depth, so this is descriptive rather than causal.

5.3 Plot 3: Gender Parity Over Time (Female and Male share by decade)

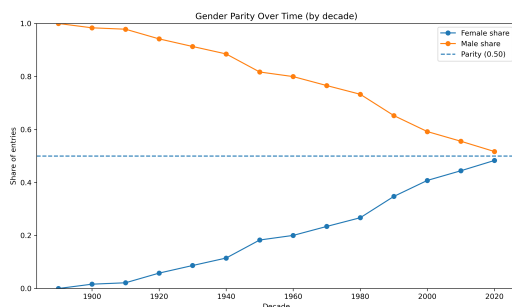


Figure 3: Female and male participation shares by decade. The dotted 0.50 line represents parity (equal female and male representation).

This plot shows a clear long-term increase in female participation over decades, moving

toward parity. It provides strong evidence that Olympic representation has become more gender-balanced over time.

Limitations: This is participation share, not medal share. Dataset composition and coverage across early editions can affect decade-level estimates.

5.4 Plot 4: Largest Decade-to-Decade Changes in Female Share

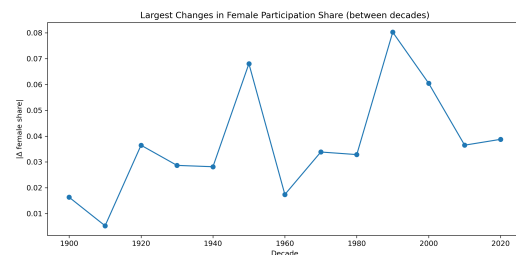


Figure 4: Absolute changes in female participation share between consecutive decades. Peaks identify decades with the largest shifts in gender representation.

This plot directly identifies the decades where gender representation changed the most. The highest peaks highlight key transition periods likely associated with expanded women's events and broader inclusion policies.

Limitations: Decades with fewer Olympic editions or uneven coverage may exaggerate changes. Male share changes mirror female share by definition ($\text{male_share} = 1 - \text{female_share}$), so female change is sufficient.

5.5 Plot 5: Overall Peak Medalist Age by Decade (Median + IQR, all sports)

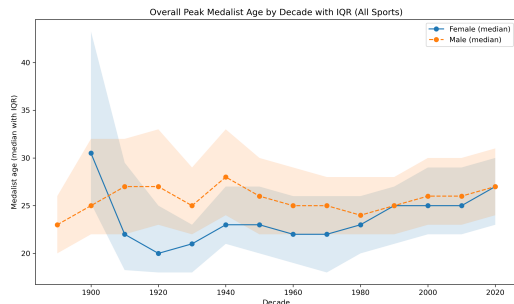


Figure 5: Median medalist age by decade with IQR shading (25th–75th percentile), computed across all sports combined and split by sex.

This figure summarizes how the typical medalist age (median) and its spread (IQR) change across decades for females and males. If medians rise, typical medal-winning ages trend older; if IQR widens, medalist ages become more varied. Early decades for women may be less stable due to smaller sample sizes.

Limitations: Combining all sports masks sport-specific peak ages. Team events can inflate medalist entries, and age imputation may slightly smooth the trend.

5.6 RQ1 Conclusion and Next Steps

Conclusion (RQ1): Overall, the results show that the Olympics become substantially more gender-balanced over time, with female participation increasing steadily and moving closer to parity in recent decades. Medalist composition varies across age groups and sex, suggesting that medal-winning patterns are not uniform when aggregated across sports. The peak-age plot indicates that typical medalist ages and their variability shift across decades, although early women’s data should be interpreted cautiously due to smaller samples.

Next steps (RQ1): With one more week, we would (1) repeat the peak-age analysis by individual sports (top sports) to avoid masking sport-specific peak ages, (2) separate Summer vs. Winter editions to compare parity and age trends by season, (3) include decade sample sizes on plots and suppress decades with very

small counts, and (4) optionally extend beyond EDA to a simple predictive model for medal probability with clear baselines (e.g., majority-class “No medal”).

(Added this section below as a sub conclusion to the EDA after placing the plots for the different research questions. I’ve added the question 1 plots you had done - Josh)

5.7 Conclusion to EDA:

Overall, the exploratory analysis shows that Olympic performance and participation are shaped by long-term national investment, increasing gender inclusion, and sport-specific differences in athlete development and peak performance age.

Common plot mistakes to avoid:

- Bar chart for a continuous variable (use histogram/density instead).
- No axis labels/units.
- Too many categories (group small ones into “Other” or filter).

6 Data Analysis (Techniques From This Course)

Pick a direction that fits your question. Use **multiple** techniques (not just one).

6.1 Analysis Plan

In 3–6 sentences: what methods did you use and why do they answer your question?

6.2 Baseline (If you do prediction)

If you predict something, include a simple baseline first (e.g., mean predictor, simple linear regression, majority class).

6.3 Main Results

Present results with at least **one table** and explain it in words.

9 Contributions (If Group Project)		
Model / Method	Metric 1 (e.g., Accuracy)	Metric 2 (e.g., RMSE)
Baseline	[]	List what each person did. Be honest and specific. []
Your method	[]	

If you used a train/test split: say how you split and why (and don't evaluate on training data only).

6.4 Interpretation (So What?)

Don't just report numbers. Explain:

- What the results mean in plain language,
- What surprised you (if anything),
- What limitations might change your conclusion.

7 Real-World Application

Your project should produce **useful insight** or a **decision** someone could act on.

Answer:

- Who is the stakeholder? (customer, city planner, student, business, etc.)
- What decision could they make based on your results?
- What would you recommend (1–3 bullets), and how confident are you?

8 Report Quality and Reproducibility

8.1 How to run your code

- Where your code is (GitHub/ZIP):
- Main file/notebook name:
- Steps to run (3–6 bullets):
- Required packages (if any):

8.2 Project structure (recommended)

- README.md (how to run)
- notebooks/ (analysis notebooks)
- src/ (helper functions, optional)
- figures/ (exported plots, optional)

- Member 1:
- Member 2:
- Member 3:

10 Conclusion and Next Steps

Conclusion: 1 paragraph summarizing the answer to your question.

Next steps: If you had 1 more week, what would you improve or explore?

11 AI Disclosure

If you used AI tools (e.g., ChatGPT, Gemini) for coding or writing, disclose it.

- Tools used (if any):
- What you used them for (debugging, rewriting, brainstorming, etc.):
- Prompts you used:

References (No research papers required)

You do **not** need research papers. You should cite:

- Your dataset link (required),
- Any important documentation/tutorial you relied on,
- Any external code sources (if any).
- Dataset: [Name], [Link], accessed [date].
- Documentation: [Library/Tool], [Link].

A Appendix (Optional)

Extra figures, extra tables, or small code snippets that would clutter the main report.

A.1 "Proportion of Olympic Medals Won by Continent"

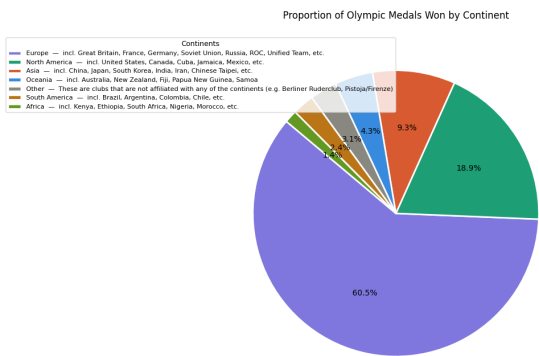


Figure 6